

Prateek Sinha

Chennai, India | ps826105@gmail.com | github.com/CzPhantom10 | +91 9142116703

SUMMARY

AI/ML enthusiast with hands-on experience in building LLMs, Retrieval-Augmented Generation (RAG) systems, and reinforcement learning environments. Proficient in Data Structures & Algorithms in Python. Skilled in fine-tuning transformer models, developing end-to-end AI applications, and deploying full-stack solutions. Strong hackathon performer focused on real-world AI systems.

EDUCATION

SRM Institute of Science and Technology

B.Tech in Computer Science and Engineering
– CGPA: 8.35

Kattankulathur, Tamil Nadu

2023 – 2027

PROJECTS

RAG Chatbot System | *FastAPI, ChromaDB, Sentence-Transformers, Next.js*

Oct 2024

- Built end-to-end RAG system for document QA achieving ~85% retrieval relevance and ~40% fewer hallucinations vs. baseline
- Designed ingestion pipeline with chunking, embedding, and ChromaDB vector search
- Developed Next.js frontend with FastAPI backend; real-time query handling under 2s response time

LLM From Scratch | *Python, PyTorch*

Mar 2025

- Implemented character-level Transformer LM from scratch with tokenization and autoregressive generation on 1M+ char corpus
- Built training pipeline with AdamW optimization and checkpointing; stable convergence within 5000 steps
- Evaluated model via perplexity tracking and text generation quality across checkpoints

QuestMint | *React, Flask, Firebase, Solidity, Groq API*

Apr 2025

- Built AI-powered web3 gaming platform with dynamic quest generation and blockchain NFT collectibles; Top 25 at HackHazard 2025
- Developed full-stack system with React, Flask, and Firebase for real-time multiplayer gameplay
- Implemented gas-optimized Solidity smart contracts for NFT minting and trading using Ethers.js

LawGPT | *Python, PyTorch, Transformers*

May 2025

- Fine-tuned GPT-2 on 50K+ Indian legal QA pairs (IPC, CPC) for domain-specific question answering
- Instruction tuning pipeline via Hugging Face Transformers; 30% faster training with gradient checkpointing
- Built inference pipeline with context-aware legal responses at avg. under 3s latency on CPU

APEX AI Survival Arena | *PyTorch, Pygame, NumPy*

Sept 2025

- Built multi-agent RL system using DQN with 8+ agents; achieved 3x survival rate over random baseline in 500 episodes
- Designed 12×12 grid environment with resource management, seasonal dynamics, and 11-action/25-dim state space
- Implemented experience replay and target network stabilization; reduced training loss ~60% vs. vanilla Q-learning

ACHIEVEMENTS

Top 25 AIR – HackHazard 2025 (Monad Track) for QuestMint

Top 100 AIR – HackHazard 2025 (Overall)

Top 10 – AMU Hacks 4.0 for ChikitchakAI

CERTIFICATIONS

GitHub Copilot – Microsoft (Apr 2026) | Verify

Machine Learning with Python (ML0101EN) – IBM / Cognitive Class (Oct 2025) | Verify

C++ For C Programmers, Part A – University of California, Santa Cruz / Coursera (Mar 2024) | Verify

TECHNICAL SKILLS

Languages: Python (DSA, AI/ML – Intermediate), SQL, HTML, CSS, JavaScript

Tools and Framework: Streamlit, Flask, Fast API, AutoCAD, Git

AI/ML: PyTorch, Pygame, Transformers, TensorFlow, LangChain, ChromaDB, OpenCV, Scikit-learn, Pandas, Numpy, Matplotlib, Seaborn

Domains: LLMs, Generative AI, Computer Vision, Database Management, NLP, Data Science, DevOps